EE-312L Signals and Systems Lab
Laboratory Session 13
To Understand and Compute the Laplace Transform and Inverse Laplace Transform using MATLAB/OCTAVE

January 1, 2025

1 Objective

The objective of this lab session is to understand and compute the Laplace Transform and Inverse Laplace Transform using MATLAB.

2 Laplace Transform of Continuous-Time Systems

The Laplace Transform of a continuous-time signal

$$X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt, \tag{1}$$

where $s = \sigma + j\omega$ The MATLAB command laplace is used to compute Laplace Transform of continuous-time systems.

Example 01:

Compute the (unilateral) Laplace transform of the function $x(t) = e^{-t}$.

```

clc;
clear all;
close all;

syms t s

x = exp(-t);
```

```
laplace(x)
```

Result: $\frac{1}{s+1}$

Example 02:

Compute the Laplace transform of the function $x(t) = \sin(\omega t)u(t)$.

```

clc;
clear all;
close all;

syms t s w

x = sin(w*t);
laplace(x)
```

Result: $\frac{\omega}{s^2+\omega^2}$

Example 03:

Compute the Laplace transform of the function $x(t) = \cos(\omega t)u(t)$.

```

clc;
clear all;
close all;

syms t s w

x = cos(w*t);
laplace(x)
```

Result: $\frac{s}{s^2+\omega^2}$

Example 04:

Compute the Laplace transform of the function $x(t) = e^{-at}\cos(\omega t)u(t)$.

```

clc;
clear all;
close all;

syms t s w a
```

```

x = exp(-a*t)*cos(w*t);
laplace(x)
```

Result: $\frac{s+a}{(s+a)^2+\omega^2}$

3 Inverse Laplace Transform

The MATLAB command ilaplace is used to compute Laplace Transform of continuous-time systems.

Example 05:

Compute the inverse Laplace transform of the function $F(s) = \frac{1}{1+s}$.

```

clc;
clear all;
close all;

syms t s

F = 1/(1+s);
ilaplace(F)
```

Result: e^{-t}

Example 06:

Compute the inverse Laplace transform of the function $F(s) = \frac{1}{s-a}$.

```

clc;
clear all;
close all;

syms t s a

F = 1/(s-a);
ilaplace(F)
```

Result: e^{at}

4 Lab 13: Lab Tasks

1. Compute the Laplace transform of the following signals using MATLAB/OCTAVE
   (a) $x_1(t) = u(t)$
   (b) $x_2(t) = e^{-at}\sin(\omega t)u(t)$
   (c) $x_3(t) = t^2 u(t)$
2. Compute the Inverse Laplace transform of the following functions using MATLAB/OCTAVE
   (a) $X_1(s) = \frac{1}{s}$
   (b) $X_2(s) = \frac{3!}{(s+a)^4}$

References